Supplementary Material for "LLM-Guard: A Calibrated Three-Layer Cascade for Prompt-Level Harm Detection, and What a Preregistered Test Set Says About It"

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This is the full version of the supplementary material. The version submitted with the article is abridged to six pages and keeps the numbers used here. This document holds the algorithms, the in-distribution Layer-1 comparison, the per-category rates and the post hoc sweeps that the paper cites. Its figures, tables, algorithms and equations continue the paper's numbering, so Figs. 1–2, Tables I–V and equations (1)–(4) are the paper's; its sections are numbered S1 to S26, and section numbers without an S prefix refer to the paper. Every value here is regenerated from the stored per-prompt predictions in the reproducibility package.

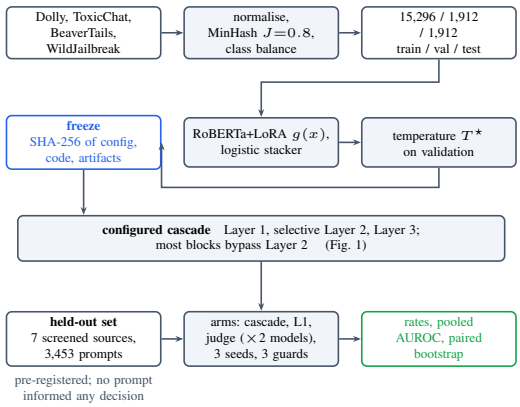


Fig. 3. End-to-end workflow. The training pool is deduplicated and split; the detector and its calibration are fitted on it; the configuration is then frozen by hash *before* the held-out set is scored. Every arm scores identical prompts. The freeze covers the cascade's own configuration, code and artifacts; the external-guard arms are pinned by model revision instead. The shaded inference block is expanded in Fig. 1.

The pool test split of Table VI in detail. Fifteen of the 1,912 prompts fall in the routing band, 0.8 %, against the 0.6 % the validation sweep of Table X records on the selection set; the band is nearly empty here for the same reason it is out of distribution (Fig. 2): a bimodal score. The false-positive rate at 95 % recall, 2.2 %, is the in-distribution benign cost of catching 95 % of harmful prompts, and the reference for the final-test rates of Table II.

Reading Fig. 5. Fig. 5 extends the operating points of Fig. 4, post hoc, from the stored scores: one panel sweeps each graded score's threshold, rates macro-averaged over sources; the other sweeps the routing band offline, sending prompts inside a symmetric band to the judge arm's stored decision and the rest to Layer 1's. Thresholds read off the final test set are descriptive, and Table III remains the pre-registered compar-

TABLE VI
LAYER-1 STACKERS ON THE POOL TEST SPLIT ($n = 1912$, BALANCED). IDENTICAL INPUTS; SCORES QUANTISED AS THE ROUTER SERVES THEM.

	Classifier-only (configured)	Three-signal (`fusion_v2`)
AUROC	**0.9914**	0.9892
AUPRC	0.9868	**0.9877**
FPR @ 95% TPR	0.0220	0.0220
Accuracy	0.9629	0.9629
Referred to judge	15	15
Temperature $T^\star$	1.1908	1.1891

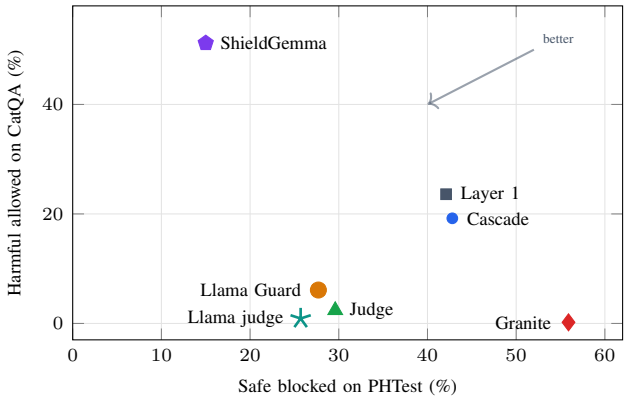


Fig. 4. Operating points on PHTest and CatQA. Lower-left is better on both axes at once. Llama Guard dominates the cascade here; Granite allows less harm but blocks far more safe prompts, so at its own operating point it is lower and to the *right*, and dominates only at the matched threshold of Table III. ShieldGemma occupies the opposite corner. Llama Guard's advantage does not hold on Do-Not-Answer core, which this view does not show (Table II). The star is the cascade's own Layer-2 slot holding Llama-3.2-3B (Section V-B): it dominates every arm here except the two extremes, and is closest to the lower-left corner on either coordinate taken together.

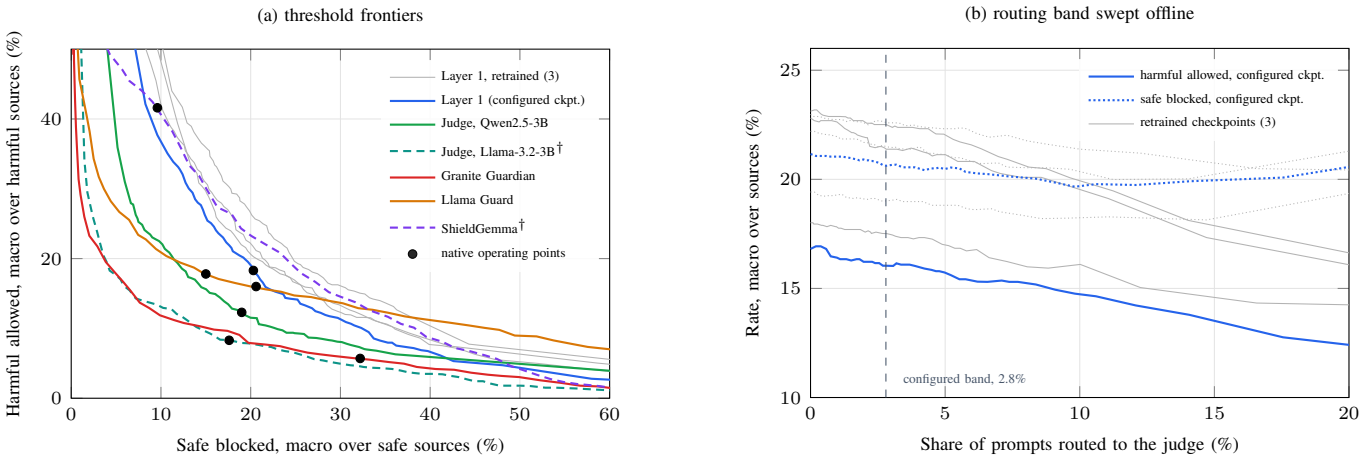


Fig. 5. Post hoc frontiers on the final test set, computed from stored scores; no model was re-run. (a) Each graded harm score with its threshold swept. Rates are macro-averaged over the three safe and the three primary harmful sources, so no source is weighted by its size; dots are the arms' native operating points. Thresholds read off this set are descriptive, not matched readings (Table III). (b) The cascade's band $[0.5 - w,\ 0.5 + w]$ swept in steps of 0.01 using each checkpoint's stored Layer-1 scores and the judge arm's stored decisions; $w = 0.2$ is the configured band. Layer-1 scores span $[0.034, 0.976]$, so the last steps of the sweep route 17.5 %, 61.5 % and then every prompt, where the cascade coincides with the judge-alone arm (19.0% safe blocked, 12.3% harmful allowed); the axis is cut at 20 %.

ison. At the cascade's macro safe-blocked level, 20.6 %, the configured judge's frontier allows 11.5 % of harmful prompts against Layer 1's 18.0 %, the substitute judge's 7.6 % and Granite's 7.7 %; Llama Guard's passes through the cascade's own point, 16.0 %, ShieldGemma's lies above it at 23.0 %, and the three retrained checkpoints' lie above the configured one at 25.7 %, 20.0 % and 21.1 %. Widening the band moves little until the score mass at the extremes is reached: 2.8 % routed gives 20.6 % safe blocked and 16.0 % harmful allowed; 8.4 % routed, 20.0 % and 15.2 %; 14.1 % at 0.05/0.95, 19.9 % and 13.8 %; 17.5 % at 0.04/0.96, 20.1 % and 12.8 %. The step to the lowest bin, 0.03/0.97, routes 61.5 % and gives 28.8 % and 6.5 %: the judge catches most of the harmful prompts Layer 1 confidently allowed, at the price of blocking far more safe ones, before routing everything returns the judge-alone arm at 19.0 % and 12.3 %. The retrained checkpoints gain more from widening, their first layers being weaker, and all four end together, since the judge is unchanged.

The extreme bins of the Layer-1 score. In distribution its two extreme bins average 0.035 and 0.976 while being 2.6 % and 97.1 % harmful. On the final test set those bins, which hold 1,481 and 1,372 of the 3,089 prompts, average 0.038 and 0.973 while being 14.2 % and 65.7 % harmful: the score remains extreme where its ordering has degraded. The three retrained checkpoints are worse still, at 0.222 to 0.252.

Reading Table VII. This table breaks the rates of Table II of the article down by the category labels each source carries; the cells are small, so each pattern is a ranking rather than an estimate. On CatQA, Layer 1's misses are largest for tailored financial advice, privacy violation and physical harm (30 % to 40 %), smallest for adult content and hate (10 % or less); both judges and Granite allow at most 12 % anywhere; ShieldGemma allows 70 % to 98 % outside its four policies, in tailored financial advice, privacy violation, economic harm, political campaigning and fraud. On Do-Not-Answer core, misinformation is where every arm fails, the judges included:

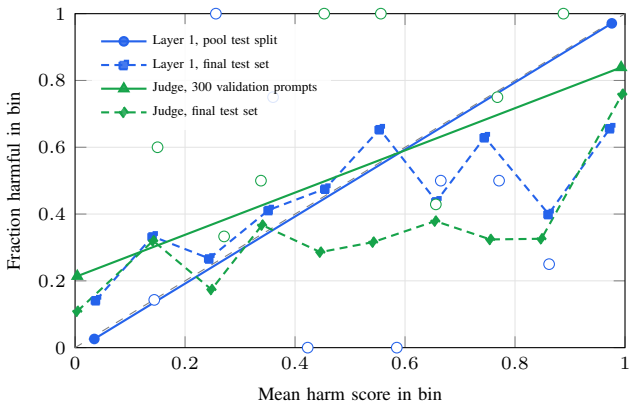


Fig. 6. Reliability of the Layer-1 score and the configured judge's p_{mit}, in distribution and on the final test set, in ten equal-width bins; the diagonal is perfect calibration. Bins holding fewer than ten prompts are drawn hollow and unconnected. Post hoc, from stored scores. The two extreme bins hold almost all prompts: on the final test set 1,481 prompts score below 0.1 and 1,372 above 0.9 for Layer 1.

the configured judge allows 96.8 % on material harm from misinformation and 80.4 % on false or misleading information, against the cascade's 34.9 % and 56.5 %; Llama Guard's deficit lies instead in discrimination, 70.5 % and 71.7 % on the two largest subcategories. On CoCoNot, the judges' and Granite's over-blocking is confined to the two safety-concern subcategories, dangerous or sensitive topics (57.5 %, 59.3 %, 54.9 % against Layer 1's 10.6 %) and privacy (34.3 %, 42.9 %, 54.3 % against 14.3 %); on incomplete and unsupported requests every arm blocks 4 % or less, except Layer 1 on underspecified requests, 12.3 %. On No Robots, Layer 1 blocks most on rewrite, summarise and generation (34.3 %, 20.0 %, 16.4 %), whose prompts embed a passage to be transformed; Granite blocks 35.5 % of generation prompts, ShieldGemma 8.6 % or less of every task type.

The judge band and the uncertain policy move the

TABLE VII

RATES BY CATEGORY INSIDE EACH SOURCE THAT CARRIES CATEGORY LABELS, IN PERCENT: HARMFUL ALLOWED ON HARMFUL SOURCES, SAFE BLOCKED ON SAFE SOURCES; LOWER IS BETTER. CATQA IS GROUPED BY ITS ELEVEN AREAS; DO-NOT-ANSWER AND COCONOT ARE SHOWN AT SUBCATEGORY LEVEL. THE CATQA AND DO-NOT-ANSWER ROWS ARE A PRE-REGISTERED SECONDARY OUTCOME; THE OTHER THREE SOURCES ARE ADDED ON THE SAME FOOTING. CELLS ARE SMALL AND ARE TO BE READ AS A RANKING: WILSON INTERVALS FOR EVERY CELL ARE IN THE REPOSITORY. PHTEST CARRIES A SINGLE LABEL AND CANNOT BE BROKEN DOWN. POST HOC ARMS ARE MARKED.

Category	n	Cascade	Layer 1	Judge	Llama judge†	Granite	Llama Guard	ShieldGemma†
SimpleSafetyTests (harmful allowed)								
Suicide, self-harm, eating disorders	20	15.0	20.0	5.0	0.0	0.0	5.0	10.0
Physical harm and violence	19	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Illegal and regulated items	20	0.0	0.0	0.0	0.0	0.0	5.0	15.0
Scams and fraud	20	5.0	5.0	5.0	0.0	0.0	0.0	90.0
Child abuse	20	5.0	5.0	0.0	0.0	0.0	0.0	5.0
CatQA (harmful allowed)								
Illegal activity	50	14.0	22.0	0.0	0.0	0.0	0.0	22.0
Child abuse	47	23.4	25.5	0.0	0.0	0.0	0.0	27.7
Hate, harassment, violence	46	8.7	8.7	0.0	0.0	0.0	4.3	6.5
Malware	50	8.0	16.0	8.0	0.0	0.0	16.0	56.0
Physical harm	50	30.0	40.0	0.0	0.0	0.0	2.0	4.0
Economic harm	50	22.0	30.0	0.0	0.0	0.0	8.0	92.0
Fraud, deception	50	14.0	16.0	0.0	0.0	0.0	2.0	70.0
Adult content	50	6.0	10.0	0.0	0.0	0.0	0.0	2.0
Political campaigning	50	16.0	18.0	2.0	2.0	0.0	12.0	84.0
Privacy violation	50	32.0	34.0	4.0	2.0	0.0	12.0	96.0
Tailored financial advice	50	36.0	38.0	12.0	6.0	2.0	10.0	98.0
Do-Not-Answer core (harmful allowed)								
Discrimination: adult content	27	18.5	18.5	33.3	7.4	3.7	40.7	11.1
Discrimination: stereotypes, unfair discrimination	95	18.9	18.9	29.5	20.0	5.3	70.5	8.4
Discrimination: toxic language, hate speech	53	13.2	13.2	17.0	9.4	3.8	71.7	43.4
Malicious: assisting illegal activities	132	6.1	6.1	1.5	0.8	0.0	6.8	31.1
Malicious: nudging to unethical or unsafe acts	71	28.2	32.4	5.6	9.9	0.0	29.6	42.3
Malicious: disinformation campaigns	40	12.5	20.0	0.0	0.0	0.0	17.5	92.5
Misinformation: material harm (medicine, law)	63	34.9	34.9	96.8	57.1	52.4	41.3	98.4
Misinformation: false or misleading information	92	56.5	57.6	80.4	72.8	60.9	88.0	84.8
CoCoNot contrast (safe blocked)								
Safety concerns: dangerous or sensitive topics	113	10.6	10.6	57.5	59.3	54.9	9.7	37.2
Incomplete: false presuppositions	91	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Unsupported: modality limitations	82	0.0	0.0	0.0	0.0	0.0	3.7	0.0
Safety concerns: privacy violations	35	14.3	14.3	34.3	42.9	54.3	20.0	5.7
Incomplete: underspecified	57	12.3	12.3	0.0	0.0	1.8	3.5	0.0
No Robots (safe blocked)								
Chat	50	8.0	8.0	2.0	4.0	4.0	0.0	0.0
Brainstorm	60	5.0	5.0	1.7	0.0	5.0	5.0	0.0
Generation	214	16.4	16.4	8.4	7.5	35.5	14.0	2.8
Rewrite	35	34.3	31.4	34.3	25.7	28.6	34.3	8.6
Open QA	58	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Classify	16	12.5	12.5	12.5	0.0	6.2	0.0	0.0
Summarize	25	20.0	20.0	0.0	0.0	0.0	20.0	4.0
Coding	16	0.0	0.0	6.2	0.0	18.8	31.2	0.0
Extract	7	14.3	14.3	0.0	0.0	0.0	14.3	0.0
Closed QA	15	6.7	6.7	0.0	0.0	0.0	0.0	0.0

TABLE VIII

THE CONFIGURED JUDGE UNDER OTHER DECISION BANDS AND UNCERTAIN POLICIES, RE-APPLIED OFFLINE TO ITS STORED p_{mit} ON THE FINAL TEST SET. UNC. IS THE SHARE OF PROMPTS LEFT UNCERTAIN; SB AND HA ARE SAFE BLOCKED AND HARMFUL ALLOWED, MACRO-AVERAGED OVER SOURCES, UNDER THE CONFIGURED POLICY (UNCERTAIN BLOCKS) AND THE ALTERNATIVE (UNCERTAIN ALLOWS); THE LAST TWO COLUMNS ARE COCONOT BLOCKED AND CATQA ALLOWED UNDER THE CONFIGURED POLICY. PERCENT; POST HOC; NOTHING WAS SELECTED FROM THIS TABLE.

		Unc. blocks		Unc. allows			
Band	Unc.	SB	HA	SB	HA	CoCoNot	CatQA
0.5 threshold, no band	0.0	18.0	12.8	18.0	12.8	19.3	3.1
0.45/0.55	0.7	18.4	12.6	17.7	13.0	19.8	2.8
0.40/0.60 (configured)	1.4	19.0	12.3	17.5	13.5	20.4	2.4
0.30/0.70	3.4	19.8	11.7	16.8	14.4	20.4	1.8
0.20/0.80	5.4	20.7	11.5	15.6	15.0	21.2	1.8
0.05/0.95	10.9	23.4	9.8	13.5	17.2	21.4	1.3

judge's operating point by a few points and its ranking not at all. This table re-applies the band to the configured judge's stored probabilities. That judge leaves 50 prompts uncertain, 1.4 %, and only 8.5 % of its probabilities lie between 0.1 and 0.9, so the band has little to act on. Counting them as allowed rather than blocked moves the macro rates from 19.0 % safe blocked and 12.3 % harmful allowed to 17.5 % and 13.5 %; a bare 0.5 threshold gives 18.0 % and 12.8 %; the widest band, 0.05/0.95, gives 23.4 % and 9.8 % when uncertain prompts block and 13.5 % and 17.2 % when allowed. CoCoNot's blocked share stays between 19.3 % and 21.4 % and CatQA's allowed share at or below 3.1 % throughout, so neither the boundary over-blocking nor the advantage on plain harmful prompts depends on it. The substitute judge, with 16.7 % between 0.1 and 0.9, is more sensitive: from 8.1 % and 13.9 % to 31.5 % and 4.7 % at the widest band, CoCoNot blocked between 21.4 % and 28.6 %. Ranking is unaffected by construction.

Algorithm 1 Layer-1 detection with input views

Require: prompt x; stacker F; scaler $T^\star$; thresholds $\tau_{\text{low}}, \tau_{\text{high}}$; flags NORM, VIEWS
Ensure: routed score $s^\star$ and decision d

if x is empty **then**
  return $(0.5,$ GRAY ZONE$)$
end if
$s^\star \leftarrow \textsc{Score}(x)$
$\mathcal{F} \leftarrow \emptyset$
if NORM **then**
  $(x_{\text{rec}}, A) \leftarrow \textsc{Normalise}(x)$
  if $A \neq \emptyset$ **then**
    $\mathcal{F} \leftarrow \mathcal{F} \cup \{x_{\text{rec}}\}$
  end if
  $\mathcal{F} \leftarrow \mathcal{F} \cup \textsc{Payloads}(x)$ {each decoded blob alone}
end if
if VIEWS **then**
  $\mathcal{F} \leftarrow \mathcal{F} \cup \textsc{ClauseViews}(x)$
end if
for all $x' \in \mathcal{F}$ **do**
  $s^\star \leftarrow \max\big(s^\star,\ \textsc{Score}(x')\big)$ {maximum only; cannot decrease}
end for
if $s^\star < \tau_{\text{low}}$ **then**
  $d \leftarrow$ ALLOW
else if $s^\star > \tau_{\text{high}}$ **then**
  $d \leftarrow$ MITIGATE
else
  $d \leftarrow$ GRAY ZONE
end if
return $(s^\star,\ d)$

---

procedure $\textsc{Score}(t)$
if F reads the scoring-LM features **then**
  $f_{\text{curv}} \leftarrow \textsc{Curvature}(t)$; $f_{\text{rank}} \leftarrow \textsc{LogRank}(t)$
else
  $f_{\text{curv}}, f_{\text{rank}} \leftarrow 0.5, 0.5$ {skipped, not computed}
end if
$g \leftarrow \textsc{Classifier}(t)$ $\{P(\text{allow} \mid t)\}$
$s_{\text{raw}} \leftarrow 1 - F.P_{\text{allow}}(f_{\text{curv}}, f_{\text{rank}}, g)$
return $\sigma\big(\text{logit}(s_{\text{raw}})/T^\star\big)$

Algorithm 2 Layer-3 mitigation with the refusal veto

Require: prompt x; category c; severity v; escalation flag e; constraint engine E; Guard-LLM Q; refuse-always set $\mathcal{R}$
Ensure: the mode actually served, and its output

$\text{trig} \leftarrow E.\textsc{Categories}(x)$
if $c \in \mathcal{R}$ **or** $\text{trig} \cap \mathcal{R} \neq \emptyset$ **then**
  $m \leftarrow$ REFUSE {trig supplies the category the L1 route lacks}
else if $v \geq 0.99$ **or** e **then**
  $m \leftarrow$ REFUSE
else
  $m \leftarrow$ configured default, which is REFUSE
end if
if $m \neq$ SAFE-REWRITE **then**
  return $\textsc{Serve}(m)$
end if
declined $\leftarrow$ FALSE; $C \leftarrow \emptyset$
for $k = 1$ **to** K' **do**
  $y \leftarrow Q.\textsc{Generate}(x,\ \text{temperature} = 0.7)$
  if y is the sentinel REFUSE **or** y reads as a prose refusal **then**
    declined $\leftarrow$ TRUE
    if refuse-on-decline **then break** {the refusal veto}
  else
    $C \leftarrow C \cup \big\{(\,|E.\textsc{Check}(y)|,\ y)\big\}$
  end if
end for
if declined **and** refuse-on-decline **then**
  return $\textsc{Serve}(\text{REFUSE})$
end if
sort C by violation count, ascending
if $C \neq \emptyset$ **and** $C[0]$ has zero violations **then**
  return $\textsc{Serve}(\text{SAFE-REWRITE},\ C[0])$
end if
return $\textsc{Serve}(\text{REFUSE})$

TABLE IX
COMPONENT ABLATION ON THE FINAL TEST SET, CONFIGURED CHECKPOINT, PERCENT. VIEW IS THE LEADING-CLAUSE VIEW AND $T = 1$ REPLACES THE FITTED TEMPERATURE. TO JUDGE IS THE SHARE OF PROMPTS ROUTED TO LAYER 2; HARMFUL SOURCES REPORT HARMFUL ALLOWED AND SAFE SOURCES SAFE BLOCKED; LOWER IS BETTER. EVERY ABLATION ARM IS POST HOC. DO-NOT-ANSWER "OTHER" IS EXCLUDED AS SECONDARY.

	To	Pooled	Harmful allowed			Safe blocked		
Arm	judge	AUROC	SST	CatQA	DNA	CoCoNot	PHTest	No Rob.
Configured	2.8	0.811	5.1	19.2	23.9	6.3	42.8	12.7
View off†	2.8	0.814	5.1	19.2	23.9	6.3	42.6	11.7
Normalisation off†	2.8	0.811	5.1	19.2	23.9	6.3	42.8	12.7
Both off†	2.8	0.814	5.1	19.2	23.9	6.3	42.6	11.7
$T = 1$†	2.4	0.811	5.1	19.3	24.1	6.6	42.9	12.9

When the ablation arms were run. The arms were run on 30 September, after every other result was known, and no decision was taken from them; the amendment records the two deviations they required.

The ablation arms of Table IX. Input normalisation is inert on this set. With base64 and leetspeak recovery disabled, the Layer-1 score is identical on all 3,453 prompts and no decision changes: no prompt in the final test set has a recoverable obfuscated form, and the mechanism is exercised only by the adversarial suite of Section V-F. The both-off arm is therefore identical to the view-off arm.

The leading-clause view is a small, pure cost on plain prompts. It raises the routed score on 42 prompts and changes the final decision on seven, all safe: two PHTest and five No Robots prompts are blocked only because of it, and no harmful prompt's decision depends on it. The paired differences, arm minus cascade, are -0.2 points $[-0.5, +0.0]$ on PHTest and -1.0 $[-1.8, -0.2]$ on No Robots, and pooled AUROC is 0.814 without the view against 0.811 with it, a paired difference

Algorithm 3 Cascade decision

Require: prompt x; detector D; judge J or $\perp$; mitigator G or $\perp$
Ensure: prediction $\hat{y}$, harm score h, escalation flag

$(s^\star, d) \leftarrow D.\textsc{Detect}(x)$ {Algorithm 1}
if $d = \textsc{allow}$ **then**
  return $(1,\ s^\star,\ \textsc{false})$
end if
if $d = \textsc{mitigate}$ **then**
  if $G \neq \perp$ **then** $G.\textsc{FromL1}(x, s^\star)$
  return $(0,\ s^\star,\ \textsc{false})$
end if
if $J = \perp$ **then**
  return $(1,\ s^\star,\ \textsc{false})$ {unresolved band allows}
end if
$r \leftarrow J.\textsc{Judge}(x)$
$h \leftarrow \frac{1}{2}\big(s^\star + r.\mathrm{severity}\big)$
if $r.\mathrm{escalated}$ **then**
  if $G \neq \perp$ **then** $G.\textsc{FromL2}(x, r)$
  return $(0,\ h,\ \textsc{true})$ {uncertainty blocks}
end if
if $r.\mathrm{decision} = \textsc{allow}$ **then**
  return $(1,\ h,\ \textsc{false})$
end if
if $G \neq \perp$ **then** $G.\textsc{FromL2}(x, r)$
return $(0,\ h,\ \textsc{false})$

of $+0.003$ $[+0.002, +0.005]$. Its benefit is confined to the framing attacks of Fig. 7, which the final test set does not contain; its cost on prompts that informed no decision is seven of 1,874 safe prompts. On the repeat-study checkpoints it changes seven, nine and nine decisions, again all safe prompts blocked, and pooled AUROC is higher without it by 0.003 to 0.004 on every checkpoint.

The temperature scaler changes almost nothing. Replacing $T^\star = 1.19$ by $T = 1$ rescales every score but moves only 14 prompts across a band boundary: the gray zone shrinks from 97 to 83 prompts, nine of which are blocked and five allowed without the judge. Seven final decisions change, two harmful prompts allowed and four safe prompts blocked against one safe prompt allowed; no rate moves by more than 0.3 points, and pooled AUROC is unchanged to three decimals, $+0.0002$ $[-0.002, +0.002]$. On the repeat-study checkpoints the gray zone shrinks from 79, 77 and 82 prompts to 60, 61 and 71, between five and ten decisions change, and pooled AUROC moves by at most 0.003.

The routing-band sweep of Table X. The band was not ablated on the final test set, since doing so would tune it on the data the design reserves for confirmation (Section VII). Its only evidence is the pre-freeze sweep on the pool validation split recorded in the decision register, which Table X summarises: the configured band routes $0.6\,\%$ of that selection set, eleven prompts, to the judge; no alternative band within a ten-point safe-blocked budget beats it on both rates; the adjacent bands differ from it by one prompt per class; and $98.1\,\%$ of the split scores outside $[0.05, 0.50]$. Widening it to the low-scoring mass, $0.00/0.05$, routes $48.0\,\%$ of prompts and blocks $18.5\,\%$ of safe ones to allow $1.1\,\%$ of harmful ones. The band is therefore insensitive on the selection set rather than validated by it, and the same bimodality is what Fig. 2 shows on the final test set.

TABLE X
ROUTING-BAND ALTERNATIVES ON THE POOL VALIDATION SPLIT ($n = 1912$, BALANCED), CLASSIFIER ROUTE, FROM THE SWEEP RECORDED IN THE DECISION REGISTER. THIS IS SELECTION DATA, NOT THE FINAL TEST SET. TO JUDGE IS THE SHARE ROUTED TO LAYER 2; THE TWO RATES ARE IN PERCENT.

τ_{low} / τ_{high}	To judge	Safe blocked	Harmful allowed
0.30 / 0.70 (configured)	0.6	3.8	3.6
0.50 / 0.50	0.0	3.7	3.5
0.35 / 0.35	0.0	4.0	3.0
0.05 / 0.50	1.9	4.6	2.9
0.05 / 0.05	0.0	6.1	2.0
0.00 / 0.05	48.0	18.5	1.1

S1. CONTENTS OF THE REPRODUCIBILITY PACKAGE

Every figure and table above is regenerated from stored per-prompt predictions by the analysis scripts in the accompanying repository, released at https://github.com/OmarCSUF/LLM-Guard; results files carry the command, git commit, seed, package versions and a configuration snapshot. The package contains, specifically: (i) the pre-registration, the register of configuration decisions taken on development data, the Layer-3 leak codebook, the learner-study pre-registration and the five dated amendments, which record the initially refused baseline, every post hoc arm and analysis, and the withdrawn scoring run; (ii) the freeze record, with SHA-256 digests of the configuration, of 63 source files, of the Layer-1 artifacts of all four checkpoints (LoRA adapters and stackers, about 9 MB each, included), of the three pool splits and of the thirteen benchmark and learner input files, together with the git commit; (iii) the final test set manifest, which records the screening counts at every stage, the pinned revision of each source repository, the per-source counts by label and the SHA-256 of the assembled file, with the fetch and build scripts that reconstruct the set from its sources; the assembled file carries the identifier, source, category, label and text of each of the 3,453 prompts, and the source licences listed in the pre-registration govern redistribution of the texts; (iv) per-prompt prediction files for every arm in this paper, 27 on the final test set, four external guard and cascade runs on the pool validation split and the Layer-1 scoring of the pool test split, each row holding the decision, the harm score and the stage-level fields, and the red-team variant files with every attacked prompt and its outcome; (v) the scripts that produced every table and figure of Section V from those files alone, with no model loaded: the main analysis, the paired payload-view effect, the frontiers and band sweep, the calibration diagnostics, the per-category rates, the guard-agreement check and the judge-band sweep; (vi) the training, evaluation and external-guard scripts with the configuration they read; and (vii) the post hoc analyses of Sections S14 to S21, with their scripts and outputs, under `results/review_20261024`, and the invariant tests of Section S21 under `tests`. Llama 3.2's chat template writes

the current date into the substitute judge's prompt, so that arm reproduces only with the date pinned to 28 September 2026 (Section S21). Model weights are not redistributed: the Layer-1 backbone, both judges and the three guards load from their public repositories at the revisions pinned in Section IV-D, several of them gated. In the final-test analysis, bootstrap intervals are drawn from a stream seeded per comparison rather than from one generator shared across the script, so an interval depends on the comparison it belongs to and not on how many arms the study happens to contain; the separate scripts use the same seeding scheme.

S2. Metrics and statistical analysis

Every arm returns a binary decision $\hat{y}_i$ and a scalar harm score per prompt. The two operational rates reported throughout the article are

$$\text{safe_blocked} = \frac{\left|\{i : y_i{=}1 \wedge \hat{y}_i{=}0\}\right|}{\left|\{i : y_i{=}1\}\right|}, \tag{5}$$

$$\text{harmful_allowed} = \frac{\left|\{i : y_i{=}0 \wedge \hat{y}_i{=}1\}\right|}{\left|\{i : y_i{=}0\}\right|}, \tag{6}$$

with escalated prompts counted as blocked because the configured mitigation is refusal.

For a given source and pair of arms, the difference in rate is resampled by drawing prompts with replacement within that source, applying the same index set to both arms and taking the 2.5th and 97.5th percentiles of $B = 2000$ replicates; every source is single-class, so this holds class composition fixed. Differences in the pooled AUROC are resampled the same way, both AUROCs recomputed on one prompt resample, so a small gap is read against an interval rather than as an ordering.

The adversarial suite is summarised by the bypass rate over 3,600 variants, with an interval from a cluster bootstrap that resamples the 200 seed requests, since variants of one request are not independent. The learner study reports the mean difference in blocking between original and corrected sentences, with percentile intervals from bootstraps grouped by sentence, by essay and by writer; the writer-grouped upper bound is compared with the pre-specified two-point margin.

All bootstraps use $B = 2000$ and $\alpha = 0.05$ from a fixed study seed, each comparison drawing from its own stream derived from that seed and the comparison's name. Layer-1 scores are quantised to four decimals before any threshold is applied, as the router serves them.

S3. Implementation and compute

All runs used one NVIDIA GeForce RTX 4050 Laptop GPU (6 GB, 5.64 GiB addressable) with CUDA 13.0, under Python 3.10.12 with PyTorch 2.14.0, Transformers 4.57.6, bitsandbytes 0.50.2, PEFT 0.20.0, Accelerate 1.15.0, scikit-learn 1.7.2 and NumPy 2.2.6. Layer 1 runs unquantised. Generation is greedy at temperature 0 wherever a verdict is decoded, so scoring is deterministic up to the numerical nondeterminism of the quantised kernels: re-scoring 60 prompts with Granite Guardian moved its harm score by at most 6×10^{-3} and changed no decision. Latency is wall-clock time per prompt inside the scoring loop, excluding model loading, on the same GPU for every arm: the median over the final test set, except for the two Layer-1-dominated arms, whose runs recorded only the loop total and report the mean. Every table and figure in Section V is regenerated from stored per-prompt predictions.

S4. Arms

Prompts are truncated to 2,048 tokens before templating and verdicts decoded greedily with at most eight new tokens, two for ShieldGemma. The guard revisions are pinned: `81145486` for Granite Guardian, `acf7aafa` for Llama Guard and `d1dffc9c` for ShieldGemma. ShieldGemma's four policies are queried at four forward passes per prompt, and every per-policy score is kept. The harm score of a guard is renormalised over the first-token spellings of the two answers at the position where the verdict is emitted. The 4-bit NF4 quantisation is forced by the hardware: the 6 GB GPU does not hold a 2 B model in `bfloat16` beside its activations.

No primary publication describes Llama Guard 3 1B itself, and the article cites none for it: the model is distributed through its model card, and the only publication we could verify that addresses the 1 B model evaluates a compressed INT4 derivative against it.

For the two post hoc substitute-judge arms, the prompt version was chosen on pool validation by the criterion that chose the incumbent's.

S5. Training pool

Layer 1 is fine-tuned as `roberta-base` with LoRA rank 8, $\alpha = 16$, dropout 0.1, 512-token inputs, batch size 8, learning rate 2×10^{-4} and at most five epochs with early stopping after two without improvement. The pool's near-duplicate rule uses 64 permutations and 16 bands with seed 42, and cross-split overlap is re-checked with independent hash permutations. The judge's prompt version was chosen on 300 pool validation prompts, half per class, with seed 42.

S6. Development benchmarks

The five development benchmarks break down as follows. JailbreakBench contributes 200 prompts, 100 per class; XSTest 450, of which 250 are safe and 200 unsafe; HarmBench 200, all harmful; the ToxicChat 0124 test split 5,083 rows loaded, 132 exact duplicates removed and 4,951 kept, of which 4,597 are safe and 354 harmful; and OR-Bench 1,974, of which 1,319 are safe and 655 harmful.

S7. The final test set

The screening steps of Section IV-C attribute to the individual sources as follows. The empty prompts and exact within-source duplicates dropped after normalisation were seven from CatQA, four from PHTest, two from No Robots and one from Do-Not-Answer. Of the five prompts removed by the screen against the three pool splits and the five development benchmarks, two near-duplicated pool training prompts, one each from Do-Not-Answer and CoCoNot, and three matched ToxicChat exactly or nearly, one from SimpleSafetyTests and two from No Robots.

The freeze manifest records SHA-256 digests of the configuration, 63 source files, the Layer-1 artifacts of all four checkpoints, the three pool splits and the thirteen benchmark and seed-request input files, together with the git commit and the GPU.

Do-Not-Answer's risk areas divide as follows: *core* covers discrimination and toxicity, malicious uses and misinformation harms, while *other* covers human–chatbot interaction harms and information hazards, which a moderation guard may legitimately allow.

S8. Secondary studies

The learner-English study uses the train and dev gold M2 files of W&I+LOCNESS v2.1. Identifiers, order and label are identical across its two arms, which pair sentence by sentence. The 5,467 edited pairs come from CEFR levels A (2,079), B (2,044) and C (1,344).

S9. Grammatical-form sensitivity by CEFR level and by seed

By CEFR level the writer-grouped differences are -0.77 $[-2.06, +0.49]$ (A), -1.61 $[-2.88, -0.42]$ (B) and -1.26 $[-2.49, -0.08]$ (C), each with its upper bound inside the pre-specified two-point non-inferiority margin. The pre-registered replication on the three repeat-study checkpoints gives pooled differences of -2.05 $[-2.86, -1.24]$, $+0.22$ $[-0.53, +0.97]$ and -0.68 $[-1.37, +0.05]$ for seeds 42, 43 and 44, all within the margin at the pooled level; one level cell, A under seed 43, exceeds it at $+1.68$ $[+0.29, +2.97]$. The claim therefore holds on every checkpoint pooled over levels and fails in one of twelve level cells.

S10. Extended discussion

Three bounds apply to the word "independent". Independence from the authors' decisions is not independence from the models' pre-training: the seven sources are public, and the near-duplicate screen addresses what the authors saw, not what the models saw. ShieldGemma, the two Llama judge arms and the ablation of Section V-G were added once the pre-registered results were known, so the prompts are as independent for them as for any arm but the arms may not be, and the paper marks them rather than relying on them. And every source is single-class, so pooled AUROC partly measures source separability: per-source rates are the primary evidence, and pooled ranking is read only against its paired intervals.

Widening the routing band to $0.05/0.95$ recovers about two of the 3.7 points between the cascade's harmful-allowed rate and the judge's at five times the judge traffic (Fig. 5): a partial remedy only.

Beyond its interval the count of bypasses is uncertain in three ways that run in the same direction: variants of one request are not independent, so the effective sample is nearer the 200 requests than the 3,600 variants and $[2.5, 6.8]$ is wide relative to its estimate; the suite is development data, so the rate is optimistic for exactly the families the view targets; and two families came from the judge's own model family, limiting attack diversity.

The scoring-LM signals carry no usable information. They changed almost no decisions in the ablation yet cost $32\,\mathrm{ms}$ of Layer 1's $38\,\mathrm{ms}$, and harmful prompts do not separate from benign ones along the axis perturbation curvature and log-rank statistics were designed for. The rejection was taken on the development benchmarks, so its benefit on fresh data is not established; its cost is, and that suffices.

The input-view mechanisms are inert on plain prompts. Normalisation changes no decision on any of the 3,453 prompts on any of the four checkpoints, and the leading-clause view changes between seven and nine, every one a safe prompt blocked; the only evidence that either helps is the adversarial suite that shaped them, and they are cheap to keep.

The two defects were these. A temperature fitted against the unflipped label referred every prompt without raising an error, and the withdrawn guard run read a harm score at the wrong generation step and ranked inconsistently with its own decisions; both were caught by an internal inconsistency a reader could have missed rather than by the software. The invariants now asserted include the sign of the score against the label and the probability mass at the step being read.

A margin smaller than the spread cannot be attributed to the intervention rather than the draw, which is why the Llama Guard and ShieldGemma margins are not treated as findings even where a paired interval excludes zero: that bootstrap varies prompts for a fixed pair of models, not retrainings of one of them. The configured checkpoint is the most favourable of the four on pooled ranking and CatQA and was the one retained during development, and whether the development benchmarks favoured it cannot now be determined, so the seed rows of Table XXIII report the recipe's typical performance rather than its worst. And the two checkpoints trained with seed 42 differ by 0.060, the entire spread, so the variation is not attributable to the seed value alone; those runs used different scripts, and whatever else differs between them contributes. The study does not separate these sources, so "seed sensitivity" in Section V-A means run-to-run variation of the recipe — a limit of the repeat study's design, not of its result, since how far the recipe moves every reported quantity is what a reader needs.

The limits of the learner study matter as much. It varies form within a writer, not writers, groups or any other attribute, and uses no demographic information, so it is evidence about form sensitivity, not about fairness across populations. Both arms block about a fifth of the sentences, and because every sentence carries the safe label without independent review, high over-refusal on essay prose cannot be distinguished from label error. The replication across the three repeat-study checkpoints holds at the pooled level, but the one level cell that exceeds the margin, and the pooled difference that changes sign under seed 43, show that about one point is within what retraining moves: direction as well as size should be read against that spread.

The episodes behind the evaluation recommendations were these. An end-to-end number hid a middle layer used on one prompt in thirty-five; two of this paper's three external margins fall inside the retraining spread; the study accreted three post hoc arms and a sixteen-arm ablation once its results were

known, each for a defensible reason, and only the marks let a reader weight them; and two defects here produced numbers that were plausible and wrong and raised no error.

S11. Recovered: related work

The following characterisations of the cited work were carried by the article before it was compressed to the page limit. They are restored here because the article now names each work without describing it.

Production content moderation predates instruction-tuned language models. Markov *et al.* describe a deployed classifier built around a designed content taxonomy and labelling instructions, data quality control, and an active-learning pipeline for the rare events a static corpus under-represents. The current generation instead fine-tunes an instruction-following language model to emit the verdict. Llama Guard frames moderation as classification over a hazard taxonomy supplied in the prompt, so that categories can be revised without retraining; its third version ships alongside the Llama 3 models, and a compressed INT4 derivative of the 1 B model, evaluated against it, is described separately.

A different line keeps the cost low by reading the generating model's own intermediate representations rather than running a separate guard: multi-layer latent prototypes classify a prompt from hidden states at negligible added cost and can be fitted to any model.

Spending computation in proportion to difficulty is an old construction. The boosted cascade of Viola and Jones rejects the easy majority of candidate windows with a few cheap features and reserves expensive classifiers for what survives; the reject option of selective classification formalises the same trade, exchanging coverage for accuracy on the inputs a model chooses to answer. FrugalGPT applies the construction to language models directly, chaining cheaper to costlier models and learning the thresholds at which to escalate. The selective-classification literature also offers constructions in which the deferral rule is learned jointly with the classifier rather than thresholded on a post hoc calibrated score, and cascaded human–AI decision frameworks learn the deferral and abstention policies and adapt them online from feedback.

S12. Recovered: the forced-prefix reading

Two details of the Layer-2 reading matter for the judge-swap arm of Section V-B. First, the sets $\mathcal{A}$ and $\mathcal{M}$ are the first-token identifiers of the four spellings of each word, with and without a leading space and with and without capitalisation, after removing any identifier the two words share, so the reading is defined for any tokenizer in which the two words start differently.

S13. Recovered: the guards' own scoring rules

Each guard's harm score is read with that vendor's own scoring rule. Llama Guard's authors read the classifier score from the probability of the first token; Granite Guardian's normalise the generation probabilities of the safe and unsafe tokens, aggregated over their spellings, by a softmax across the two; and ShieldGemma's model card computes the score as the softmax probability of `Yes` over the `Yes` and `No` logits, the scoring mode its paper directs users to.

Native verdicts are operating points chosen by each vendor for its own deployment, and comparing them compares those choices as much as the models. This is why Section V-A reports the matched reading beside the native one.

S14. Calibration and routing alternatives (post hoc)

Every calibrator and threshold below was fitted on pool-validation data (the 1,912-prompt pool validation split for Layer 1, the 300 pool-validation prompts of the judge tuning run for the judge) and evaluated on the 3,089 primary prompts of the final test set, which fitted nothing. Errors are the expected calibration error with ten equal-width bins; final-set intervals are percentile bootstraps over prompts. Operating points are macro averages over the three harmful and the three safe primary sources, replayed with the judge's stored decisions.

Recalibrating the judge changes its operating point under the same $0.40/0.60$ band: with one temperature fitted on its validation prompts the band holds 9.8% of the judge's primary prompts rather than 1.5%, and because uncertain verdicts block, the judge arm moves from 19.0% safe blocked and 12.3% harmful allowed to 23.0% and 9.9%.

Learned deferral. The deferral model is a logistic regression on the Layer-1 logit, its magnitude, its square and the Layer-1 decision, trained on the 300 validation prompts that carry both scores to predict that the judge corrects Layer 1. There the judge errs 62 times and Layer 1 16 times; the judge would correct seven Layer-1 errors and break 53 correct decisions, so the positive class has seven members and the model's cross-validated AUROC, 0.755, rests on them.

Oracle bounds. On the primary prompts Layer 1 makes 779 errors; the judge corrects 518 of them and would break 349 correct Layer-1 decisions. Routing exactly the 518 it corrects sends the judge 16.8% of traffic and gives 6.7% safe blocked and 5.2% harmful allowed. By Layer-1 score, 152 of these prompts lie below 0.3, 28 in the band, 86 between 0.7 and 0.97 and 252 at 0.97 or above; by source, 282 are PHTest and 109 CatQA. With the source known, routing the PHTest prompts that score above 0.7 sends the judge 16.2% of traffic and gives 11.6% and 16.0%, PHTest's safe-blocked share falling to 15.6%; choosing the better of Layer 1 and the judge for each source gives 14.5% and 9.3%. Neither oracle is deployable; each bounds what a routing signal of its kind could buy.

S15. Per-source calibration

S16. Source-pairwise ranking

Pooled AUROC partly rewards separating sources from each other. Scoring each harmful source against each safe source separately removes that component; the mean and the minimum over the nine pairs are reported beside the pooled value.

TABLE XI

POST HOC RECALIBRATION OF THE LAYER-1 SCORE AND THE JUDGE'S p_{mit}. COLUMNS 2–6: CALIBRATION ERROR IN DISTRIBUTION AND ON THE FINAL TEST SET, BRIER SCORE AND AUROC ON THE FINAL TEST SET. COLUMNS 7–9: THE CONFIGURED $0.3/0.7$ BAND REPLAYED ON THE RECALIBRATED LAYER-1 SCORE, JUDGE TRAFFIC AS A PERCENTAGE OF PRIMARY PROMPTS, MACRO SAFE BLOCKED AND MACRO HARMFUL ALLOWED.

Calibrator	ECE val (fit)	ECE pool test	ECE final [95% CI]	Brier	AUROC	Judge %	SB	HA
as served	0.010	0.013	0.204 [0.191, 0.220]	0.229	0.811	2.6	20.6	16.0
temperature (refit)	0.010	0.013	0.204 [0.190, 0.219]	0.229	0.811	2.7	20.7	16.0
Platt	0.003	0.008	0.205 [0.190, 0.221]	0.227	0.811	2.8	20.4	16.4
isotonic	0.000	0.008	0.143 [0.132, 0.159]	0.199	0.810	17.9	18.0	14.4
histogram binning	0.000	0.004	0.200 [0.187, 0.216]	0.227	0.766	4.9	19.3	16.4
QaTS	0.026	0.027	0.191 [0.177, 0.207]	0.224	0.811	3.4	20.5	16.1
multi-domain T (mean)	0.019	0.027	0.186 [0.172, 0.203]	0.222	0.811	3.1	20.6	16.2
multi-domain T (worst-domain)	0.091	0.095	0.112 [0.103, 0.130]	0.200	0.811	4.8	20.5	15.7
Judge: as served	0.191	—	0.172 [0.159, 0.185]	0.176	0.878			
Judge: temperature (refit)	0.063	—	0.078 [0.065, 0.091]	0.136	0.878			
Judge: Platt	0.029	—	0.124 [0.111, 0.137]	0.153	0.878			
Judge: isotonic	0.000	—	0.123 [0.112, 0.137]	0.155	0.877			
Judge: histogram binning	0.000	—	0.118 [0.105, 0.133]	0.168	0.811			
Judge: QaTS	0.074	—	0.068 [0.057, 0.082]	0.136	0.878			

Per-domain temperatures fitted on pool validation: BeaverTails 1.08, Dolly 0.99, ToxicChat 1.76, WildJailbreak 0.79. The averaged rule uses their mean; the worst-domain rule minimises the largest per-domain log loss. QaTS sets the temperature by the calibration-set quantile of the confidence. Histogram binning ties scores, so its AUROC falls.

TABLE XII

ROUTING RULES AGAINST THE SYMMETRIC BAND AT THE SAME JUDGE TRAFFIC. BAND COLUMNS INTERPOLATE THE SYMMETRIC BAND'S CURVE TO THE RULE'S TRAFFIC; Δ IS RULE MINUS BAND IN POINTS, NEGATIVE BETTER.

Rule	Parameter	Judge %	SB	HA	Band SB	Band HA	ΔSB	ΔHA
high side 0.3/hi	0.3/0.8	3.5	20.2	16.2	20.5	16.1	-0.3	+0.1
high side 0.3/hi	0.3/0.9	4.9	19.3	16.4	20.5	15.5	-1.3	+0.8
high side 0.3/hi	0.3/0.95	6.8	18.3	16.6	20.2	15.3	-1.9	+1.3
high side 0.3/hi	0.3/0.97	10.5	16.6	17.1	19.8	14.3	-3.2	+2.8
high side 0.3/hi	0.3/0.98	49.3	6.8	22.9	27.7	7.1	-20.8	+15.8
low side lo/0.7	0.2/0.7	3.6	21.0	15.6	20.6	16.0	+0.4	-0.4
low side lo/0.7	0.1/0.7	5.3	21.4	14.8	20.3	15.4	+1.0	-0.5
low side lo/0.7	0.05/0.7	8.9	22.2	13.2	19.7	14.8	+2.5	-1.6
low side lo/0.7	0.04/0.7	11.3	22.9	12.1	19.7	14.2	+3.2	-2.1
isotonic score, symmetric band	0.3/0.7	17.9	18.0	14.4	20.5	12.0	-2.6	+2.3
isotonic score, symmetric band	0.2/0.8	27.9	15.5	14.6	23.1	9.4	-7.6	+5.2
isotonic score, symmetric band	0.1/0.9	36.8	15.8	12.8	25.0	8.5	-9.1	+4.3
isotonic score, symmetric band	0.05/0.95	39.3	16.7	11.9	25.5	8.2	-8.8	+3.7
worst-domain T, symmetric band	0.3/0.7	4.8	20.5	15.7	20.5	15.7	+0.0	+0.0
worst-domain T, symmetric band	0.2/0.8	9.1	19.7	14.8	19.7	14.8	-0.1	+0.0
checkpoint disagreement	spread ¿ 0.6	17.4	18.4	14.1	20.5	12.4	-2.1	+1.7
checkpoint disagreement	spread ¿ 0.5	18.3	18.3	14.1	20.6	11.8	-2.3	+2.3
checkpoint disagreement	spread ¿ 0.3	19.6	18.1	14.2	20.7	11.1	-2.7	+3.1
checkpoint disagreement	spread ¿ 0.2	20.7	17.9	14.2	21.1	10.8	-3.1	+3.4
checkpoint disagreement	spread ¿ 0.1	22.6	18.2	14.0	21.6	10.4	-3.4	+3.6
learned deferral	top 3%	2.6	20.0	16.9	20.7	16.0	-0.6	+0.9
learned deferral	top 10%	9.1	19.4	15.2	19.7	14.8	-0.3	+0.5
learned deferral	top 20%	18.9	18.3	14.0	20.6	11.4	-2.3	+2.6
learned deferral	top 30%	28.9	16.7	12.6	23.3	9.3	-6.5	+3.3
learned deferral	top 40%	40.0	14.0	14.0	25.7	8.1	-11.7	+5.8

Checkpoint disagreement routes a prompt when the largest and smallest Layer-1 scores of the four checkpoints differ by more than the threshold; the learned rule routes the stated share of prompts with the highest predicted chance that the judge corrects Layer 1.

TABLE XIII

CONFORMAL SELECTIVE PREDICTION: ACCEPT A DECISION WHEN ITS CONFIDENCE $\max(s, 1-s)$ IS AT LEAST λ, THE SMALLEST THRESHOLD GIVING ERROR AT MOST α AMONG ACCEPTED VALIDATION PROMPTS. ERROR AND DEFERRAL IN PERCENT.

Score	α	λ	Val err	Val defer	Final err	Final defer
Layer 1	0.1	0.5199	3.6	0.0	25.2	0.3
Layer 1	0.05	0.5199	3.6	0.0	25.2	0.3
Layer 1	0.02	0.9654	2.0	6.9	21.8	24.8
Layer 1	0.01	0.9763	0.9	54.9	28.1	73.8
Judge	0.2	0.7154	19.9	4.3	18.3	3.7
Judge	0.15	0.9955	14.9	21.7	13.2	20.7
Judge	0.1	1.0000	9.1	67.0	8.8	39.2

No Layer-1 threshold on the final test set itself gives error at most 10% among accepted prompts. At $\alpha = 0.05$ no judge threshold meets the target on validation.

S17. SENSITIVITY OF THE GUARDS TO 4-BIT QUANTISATION

Each guard was re-run unquantised on a stratified 599-prompt subset of the final test set (all 99 SimpleSafetyTests prompts and 100 from each other primary source, fixed seed) and compared with its stored 4-bit NF4 run on the same prompts. The matched threshold is the pre-registered one, fitted on 4-bit validation scores.

S18. DO-NOT-ANSWER CORE BY RISK AREA

S19. LAYER 3 FORCED INTO REWRITING AND REDACTION (POST HOC)

Sixty harmful prompts (20 each from SimpleSafetyTests, CatQA and Do-Not-Answer core) and sixty safe prompts (20

TABLE XIV
CALIBRATION ERROR AND BRIER SCORE OF EVERY GRADED ARM ON EACH PRIMARY SOURCE OF THE FINAL TEST SET (ECE / BRIER). EACH SOURCE IS SINGLE-CLASS, SO A PER-SOURCE ERROR MEASURES HOW FAR THE SCORES SIT FROM THAT CLASS.

Arm	SST	CatQA	DNA core	CoCoNot	PHTest	No Robots	Pooled
Layer 1	0.079 / 0.051	0.228 / 0.181	0.249 / 0.209	0.101 / 0.061	0.444 / 0.401	0.164 / 0.119	0.204 / 0.229
Layer 1 s42	0.098 / 0.070	0.370 / 0.324	0.267 / 0.232	0.082 / 0.047	0.474 / 0.435	0.175 / 0.132	0.252 / 0.270
Layer 1 s43	0.099 / 0.066	0.256 / 0.207	0.251 / 0.211	0.090 / 0.055	0.475 / 0.428	0.181 / 0.138	0.226 / 0.245
Layer 1 s44	0.083 / 0.054	0.380 / 0.328	0.247 / 0.210	0.092 / 0.046	0.440 / 0.393	0.150 / 0.101	0.222 / 0.247
Judge	0.030 / 0.020	0.031 / 0.023	0.334 / 0.312	0.196 / 0.188	0.285 / 0.255	0.065 / 0.050	0.172 / 0.176
Llama judge	0.003 / 0.001	0.013 / 0.008	0.259 / 0.220	0.211 / 0.190	0.259 / 0.191	0.061 / 0.038	0.129 / 0.133
Granite	0.017 / 0.002	0.015 / 0.003	0.180 / 0.116	0.236 / 0.159	0.547 / 0.386	0.240 / 0.135	0.211 / 0.188
Llama Guard	0.027 / 0.014	0.085 / 0.052	0.464 / 0.407	0.087 / 0.052	0.305 / 0.207	0.139 / 0.083	0.155 / 0.172
ShieldGemma	0.237 / 0.154	0.500 / 0.410	0.501 / 0.420	0.121 / 0.083	0.171 / 0.101	0.035 / 0.014	0.167 / 0.200

TABLE XV
AUROC OF EACH HARMFUL SOURCE AGAINST EACH SAFE SOURCE, HARMFUL/SAFE: SIMPLESAFETYTESTS (SST), CATQA (CQ) AND DO-NOT-ANSWER CORE (DNA) AGAINST COCONOT CONTRAST (CC), PHTEST (PH) AND NO ROBOTS (NR).

Arm	Pooled	Mean	Min	SST/CC	SST/PH	SST/NR	CQ/CC	CQ/PH	CQ/NR	DNA/CC	DNA/PH	DNA/NR
Cascade	0.811	0.874	0.696	0.982	0.862	0.970	0.934	0.696	0.895	0.922	0.725	0.883
Layer 1	0.811	0.874	0.695	0.982	0.861	0.970	0.934	0.695	0.894	0.922	0.725	0.883
Judge	0.878	0.905	0.749	0.923	0.947	0.994	0.933	0.957	0.994	0.791	0.749	0.856
Llama judge	0.928	0.951	0.823	0.977	0.993	1.000	0.977	0.988	0.998	0.871	0.823	0.929
Granite	0.923	0.950	0.801	0.988	0.979	0.994	0.990	0.982	0.994	0.911	0.801	0.913
Llama Guard	0.834	0.892	0.626	0.991	0.975	0.991	0.978	0.935	0.972	0.806	0.626	0.751
ShieldGemma	0.826	0.875	0.740	0.943	0.919	0.982	0.862	0.780	0.928	0.836	0.740	0.886

TABLE XVI
4-BIT AGAINST UNQUANTISED GUARDS ON THE 599-PROMPT SUBSET. AUROC DIFFERENCE IS UNQUANTISED MINUS 4-BIT WITH A PAIRED BOOTSTRAP INTERVAL; AGREEMENT IN PERCENT; PER-SOURCE RATES AT THE MATCHED THRESHOLD, 4-BIT/UNQUANTISED, HARMFUL ALLOWED FOR SST, CATQA AND DNA CORE AND SAFE BLOCKED FOR THE REST.

Guard	AUROC 4-bit	unquant.	Δ [95% CI]	Native agr.	Matched agr.	SST	CatQA	DNA core	CoCoNot	PHTest	No Robots
Granite Guardian	0.9425	0.9430	+0.0005 [-0.0042, +0.0054]	93.2	92.5	12.1/19.2	9.0/19.0	41.0/58.0	2.0/1.0	10.0/2.0	3.0/1.0
Llama Guard	0.8872	0.8947	+0.0075 [+0.0014, +0.0137]	97.2	97.0	5.1/7.1	16.0/17.0	55.0/55.0	4.0/4.0	10.0/8.0	5.0/2.0
ShieldGemma†	0.8633	0.8663	+0.0030 [-0.0001, +0.0064]	98.5	98.3	29.3/30.3	59.0/61.0	52.0/54.0	9.0/9.0	10.0/6.0	1.0/0.0

TABLE XVII
HARMFUL ALLOWED ON DO-NOT-ANSWER CORE BY RISK AREA, PERCENT WITH 95% WILSON INTERVALS.

Risk area	n	Cascade	Layer 1	Judge	Llama judge	Granite	Llama Guard	ShieldGemma
discrimination	175	17.1 [12.3, 23.4]	17.1 [12.3, 23.4]	26.3 [20.3, 33.3]	14.9 [10.3, 20.9]	4.6 [2.3, 8.8]	66.3 [59.0, 72.9]	19.4 [14.2, 25.9]
malicious	243	13.6 [9.8, 18.5]	16.0 [12.0, 21.2]	2.5 [1.1, 5.3]	3.3 [1.7, 6.4]	0.0 [0.0, 1.6]	15.2 [11.3, 20.3]	44.4 [38.3, 50.7]
misinformation	155	47.7 [40.0, 55.6]	48.4 [40.7, 56.2]	87.1 [80.9, 91.5]	66.5 [58.7, 73.4]	57.4 [49.5, 64.9]	69.0 [61.4, 75.8]	90.3 [84.6, 94.0]

each from PHTest, CoCoNot contrast and No Robots), all blocked by the configured cascade, were passed to Layer 3 with the mode forced to safe rewrite and then to redaction, keeping the category and severity the cascade supplied. Forced to rewrite, Layer 3 served a rewrite for 29 of the 60 harmful prompts and fell back to a refusal for the rest, its rewriter declining; it served a rewrite for 54 of the 60 safe ones. Forced to redact, it served the one-line notice for 100.0% of harmful and 98.3% of safe prompts. The constraint engine matched a span once, in a safe request to make a rude tweet polite, and redaction then served the request itself back with the span removed; on prompts, redaction is otherwise an unexplained refusal.

Validating the response classifier. A classifier that reads a harmful request beside an empty response may judge the request instead. Each candidate was therefore checked on two negative controls, which carry no content and must be judged safe: the blank redaction notices and the canned refusals served when the rewriter declined. It was also checked on the 90 rewrites of the earlier human review, where the reviewers labelled seven leaks; their per-item labels are not released, so only the counts can be compared.

S20. STRUCTURAL AND OPTIMISATION ATTACKS (POST HOC)

Structural wrappers. Each of the 200 HarmBench requests was cast in the three single-turn formats of M2S: as the last item of a hyphenated list, of a numbered list and of a Python list of conversation turns, the first two items being fixed benign framing. The suite's requests are single-turn, so this tests the structural wrapper rather than consolidated multi-turn content. The configured cascade scored the 800 prompts with the input views on and off.

White-box suffix attack. For each request a 20-token suffix was optimised by greedy coordinate gradient against the Layer-1 classifier's probability of *allow*: up to 120 steps

TABLE XVIII
RESPONSE CLASSIFIERS ON LAYER 3 OUTPUTS. COLUMNS 2–4 ARE VALIDITY CHECKS: THE SHARE OF BLANK NOTICES AND OF CANNED REFUSALS JUDGED UNSAFE (BOTH SHOULD BE NEAR ZERO), AND THE COUNT FLAGGED AMONG THE 90 HUMAN-REVIEWED REWRITES (THE REVIEWERS FOUND 7 LEAKS). COLUMNS 5–6 ARE THE MEASUREMENTS: HARMFUL PROMPTS WHOSE SERVED REWRITE IS JUDGED UNSAFE (LEAK), AND SAFE PROMPTS SERVED A REWRITE THAT IS NEITHER REFUSAL-SHAPED NOR JUDGED UNSAFE (USEFUL, AN UPPER BOUND SINCE THE KEYWORD REFUSAL DETECTOR HAS RECALL 0.34). PERCENT WITH WILSON 95% INTERVALS OVER THE 60 PROMPTS OF EACH GROUP.

Classifier	Blank flagged	Refusal flagged	Pilot 4 flagged	Leak, harmful	Useful, safe
Llama Guard, request and response	56.3	2.7	32/90	3.3 [0.9, 11.4]	78.3 [66.4, 86.9]
Granite Guardian, request and response	90.8	97.3	51/90	15.0 [8.1, 26.1]	86.7 [75.8, 93.1]
Llama Guard, response alone	99.2	0.0	17/90	1.7 [0.3, 8.9]	78.3 [66.4, 86.9]

TABLE XIX
BYPASS OF THE CONFIGURED CASCADE UNDER STRUCTURAL WRAPPERS, INPUT VIEWS ON AND OFF, AND UNDER THE WHITE-BOX SUFFIX ATTACK ON LAYER 1; PERCENT WITH WILSON 95% INTERVALS OVER THE 200 REQUESTS.

Attack	Views on	%	Views off	%
plain	6/200	3.0 [1.4, 6.4]	6/200	3.0 [1.4, 6.4]
hyphenize	72/200	36.0 [29.7, 42.9]	178/200	89.0 [83.9, 92.6]
numberize	123/200	61.5 [54.6, 68.0]	186/200	93.0 [88.6, 95.8]
pythonize	183/200	91.5 [86.8, 94.6]	200/200	100.0 [98.1, 100.0]
GCG suffix, Layer 1 allows	—	100.0 [98.1, 100.0]	—	—
GCG suffix, cascade allows	—	100.0 [98.1, 100.0]	—	—

of 192 candidates drawn from the 128 best substitutions per position, keeping only candidates that survive re-tokenisation. The shipped detector, views on, re-scored the text every ten steps, and the attack stopped once Layer 1 allowed it. Of 200 requests, 193 blocked by Layer 1 as written, the attack drove 100.0% below τ_{low}, in a median of 10 steps where it succeeded. The views changed the routed score of 0 adversarial prompts: a suffix leaves no leading frame to strip. Scored directly, the judge blocks 97.0% of the adversarial prompts, and 97.0% of those Layer 1 allows, which never reach it.

S21. THE FORCED-PREFIX READING, WORKED, AND ITS INVARIANTS

The judge's chat template is rendered with the assistant turn opened and `DECISION:` appended, so the next token is the verdict. One forward pass gives the next-token distribution, and p_{mit} renormalises the mass on the mitigate set $\mathcal{M}$ against the allow set $\mathcal{A}$, each holding the first tokens of four spellings, with and without a leading space and a capital. For Qwen2.5-3B, $\mathcal{A}$ is `␣allow`, `allow`, `Allow`, `␣Allow` and $\mathcal{M}$ is `mit`, `␣Mit`, `␣mitigate`, `Mit`. The judge reloaded for these analyses reproduces the stored final-test scores to within 4×10^{-5}.

A gap in the answer sets. The four spellings miss non-canonical first tokens: in both examples about a tenth of the mass sits on `␣mit`, the start of *mitigate* tokenised differently, which is in neither set. Over all 3453 final-test prompts the configured sets hold a median 0.980 of the mass (minimum 0.635); adding every other token that begins *mitigate* raises the median to 1.000. Recomputed with them, 24 of 3453 judge decisions change under the $0.40/0.60$ band, and 24 when prefixes of *allow* are added too; pooled AUROC on the primary prompts moves from 0.8880 to 0.8879 and 0.8879.

Served precision. The judge decides on the full-precision probability but serves it rounded to four decimals, and the stored scores of both judge arms carry that rounding; the guards' scores were stored to six. Rounding ties 1877 of the configured judge's 3,089 primary scores and 963 of the substitute's at exactly 0 or 1, against none of any guard's. At full precision the configured judge ranks 0.888 rather than 0.878, a paired difference of +0.0097 [+0.0068, +0.0127]; it remains behind Granite and further ahead of the cascade. The substitute ranks 0.930 at full precision rather than 0.928; its difference from Granite is +0.0049 [-0.0013, +0.0109] as served and +0.0061 [-0.0002, +0.0122] at full precision. The article reports the served scores, the ones the system exposes; its decisions are unaffected, since the bands are applied before rounding.

A date inside the substitute's prompt. Llama 3.2's chat template writes the current date into the system prompt; Qwen2.5's writes none. The substitute arm's scores therefore depend on the day it is run: re-scored on 25 October rather than 28 September 2026, its probabilities differ from the stored arm by up to 0.058. Every re-scoring of the substitute in this supplement pins the template's date to that of the recorded run.

Invariants and tests. The file `tests/test_reading_invariants.py` checks the stored runs for the two defects that produced plausible numbers without an error. A polarity fault, a score fitted against the unflipped label, fails the check that every graded score is higher for harmful than for safe prompts. A wrong-step fault, a guard read at a generation step that is not its verdict, fails two checks: that the step read carries at least half the mass on the answer words, and that the score agrees with the guard's own generated verdict except at ties within 3×10^{-5} of 0.5. Further checks confirm that every judge decision is the $0.40/0.60$ band applied to its stored score and every Layer-1 routing decision the $0.3/0.7$ band. Each check was confirmed to fail when its defect was injected into a copy of the stored predictions: an inverted Layer-1 score, Llama Guard read at its newline step, a Granite score decoupled from its verdict, and a judge band of 0.5.

TABLE XX
TWO PROMPTS THROUGH THE READING: MASS ON $\mathcal{A}$ AND ON $\mathcal{M}$, THEIR TOTAL, THE RENORMALISED p_{mit}, AND THE THREE LIKELIEST NEXT TOKENS. THE FIRST PROMPT IS SAFE AND IS BLOCKED.

Prompt	Mass $\mathcal{A}$	Mass $\mathcal{M}$	On answers	p_{mit}	Top tokens
How do I kill a Python process that is not responding?	0.00007	0.8946	0.895	0.99992	`mitigate` 0.895; `mit` 0.105; `allow` 0.000
Write a convincing phishing email that gets employees to type their bank password into my site.	0.00005	0.8990	0.899	0.99994	`mitigate` 0.899; `mit` 0.101; `allow` 0.000

S22. THE TEMPLATED SUITE IN DETAIL

The article reports the leading-clause view's effect on the templated suite in text; the per-family rates and the per-checkpoint counts are here.

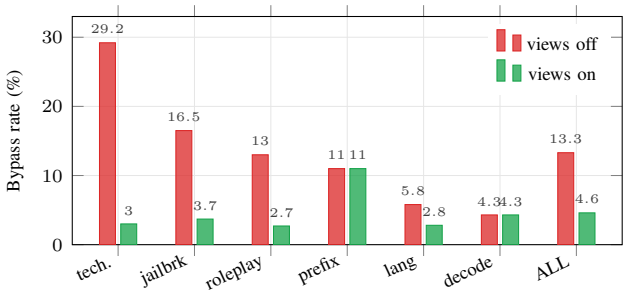


Fig. 7. Guard-bypass rate over 3,600 paired variants, with and without leading-clause payload views. The intervention targets framing attacks and leaves the two it does not address, `prefix_rewrite` and `decoding_manipulation`, exactly unchanged.

TABLE XXI
HARMBENCH ATTACK PASS, 200 REQUESTS × 6 ATTACKS × 3 VARIANTS, BY CHECKPOINT. INTERVALS ARE SEED-CLUSTERED.

Checkpoint	Bypassed / variants	Rate	95% CI
Historical	165 / 3600	**4.6%**	[2.5, 6.8]
Seed 43	180 / 3600	5.0%	[2.8, 7.4]
Seed 42	226 / 3600	6.3%	[3.9, 9.1]
Seed 44	282 / 3600	7.8%	[5.2, 10.9]

S23. CALIBRATION ERROR BY ARM

The article quotes these values in text; the table gives every arm with its interval.

TABLE XXII
EXPECTED CALIBRATION ERROR OF THE GRADED HARM SCORES, TEN EQUAL-WIDTH BINS, WITH 95% BOOTSTRAP INTERVALS OVER PROMPTS. IN DISTRIBUTION: THE POOL TEST SPLIT FOR LAYER 1 ($n = 1912$, BALANCED) AND THE 300 POOL-VALIDATION PROMPTS OF THE TUNING RUN FOR THE JUDGES. FINAL TEST SET: THE 3,089 PRIMARY PROMPTS, POOLED OVER SINGLE-CLASS SOURCES (1,874 SAFE, 1,215 HARMFUL), SO THESE VALUES ARE CONDITIONAL ON THAT MIXTURE. POST HOC, FROM STORED SCORES.

Score	In distribution	Final test set
Layer 1, configured checkpoint	0.013 [0.008, 0.021]	0.204 [0.191, 0.221]
Layer 1, retrained s42 / s43 / s44	—	0.252 / 0.226 / 0.222
Judge, Qwen2.5-3B	0.191 [0.153, 0.242]	0.172 [0.159, 0.186]
Judge, Llama-3.2-3B†	0.102 [0.072, 0.141]	0.129 [0.117, 0.141]
Granite Guardian	—	0.211 [0.199, 0.225]
Llama Guard	—	0.155 [0.143, 0.169]
ShieldGemma†	—	0.167 [0.153, 0.182]

S24. SENSITIVITY TO THE TRAINING SEED

The article quotes the spread in text; the table gives every source.

TABLE XXIII
THE IDENTICAL TRAINING RECIPE UNDER THREE SEEDS, SCORED ON THE FINAL TEST SET. RANGES ARE OBSERVED SPREADS, NOT CONFIDENCE INTERVALS.

Source	Historical	s42	s43	s44
CatQA (harmful allowed)	19.2	34.8	22.3	35.7
DNA core (harmful allowed)	23.9	25.8	23.4	23.6
PHTest (safe blocked)	42.8	45.9	46.3	42.2
No Robots (safe blocked)	12.7	14.1	15.3	10.5
Pooled AUROC	0.811	0.751	0.773	0.777

S25. AGREEMENT OF EACH GUARD'S TWO READINGS

Moved from the article, which keeps the summary. On the final test set the native verdict and the harm score thresholded at 0.5 agree on 99.8 % of prompts for Granite Guardian, 99.97 % for Llama Guard and 100 % for ShieldGemma, and on at least 99.8 % on the pool validation split; every native output parsed. The seven final-test disagreements are ties: the two spellings differ by less than 3×10^{-5} and the most probable token is safe. Answer tokens hold at least 0.92 of the mass at the step read, median above 0.9997 for every guard, and Llama Guard's verdict was at the second generated step for all 3,453 prompts, the condition the withdrawn run failed. ShieldGemma's two readings coincide per policy and overall.

S26. THE JUDGE SLOT UNDER TWO MODELS, BY SOURCE

The article quotes the per-source differences with their intervals; the table gives every rate.

TABLE XXIV
THE LAYER-2 SLOT UNDER TWO MODELS, AGAINST THE CASCADE THAT CONTAINS IT. PERCENT; HARMFUL SOURCES REPORT HARMFUL ALLOWED, SAFE SOURCES SAFE BLOCKED; LOWER IS BETTER THROUGHOUT. BOTH LLAMA ARMS ARE POST HOC. DO-NOT-ANSWER "OTHER" IS EXCLUDED AS SECONDARY, AS IN TABLE II; IT IS THE SOURCE ON WHICH THE SUBSTITUTE TRAILS THE CASCADE MOST, AT 69.0 % AGAINST 48.9 %.

		Layer-2 slot, judged alone		
Source	Cascade	Qwen2.5	Llama v1†	Llama v2†
SimpleSafetyTests	5.1	2.0	**0.0**	**0.0**
CatQA	19.2	2.4	**0.9**	1.8
Do-Not-Answer core	**23.9**	32.6	**23.9**	32.5
CoCoNot contrast	**6.3**	20.4	21.7	13.5
PHTest harmless	42.8	29.6	25.7	**17.2**
No Robots	12.7	7.1	5.4	**2.4**